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Solenoid valve

Abstract

A solenoid valve includes a water inlet valve seat, a guide seat, a coil barrel, and a water outlet valve seat. When in an open state, an iron core is moved to close a pressure relief hole via an iron core spring and water pressure, and the diaphragm blocks the water flow of a water inlet via water pressure and the force of a guide spring, so as to achieve a bistable effect. The coil barrel includes two pairs of magnets and a pair of magnetically conductive sheets to form two magnetic routes. The solenoid valve has a simple structure and can be produced easily.

Inventors:	Ma; Cheng-Yi (Nantou Hsien, TW)
Applicant:	HYDROTEK CORPORATION (Nantou Hsien, TW)
Family ID:	91192933
Assignee:	HYDROTEK CORPORATION (Nantou, TW)
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Primary Examiner: Jellett; Matthew W

Attorney, Agent or Firm: Sinorica International Patent & Trademark

Background/Summary

FIELD OF THE INVENTION

(1) The present invention relates to a solenoid valve for a fluid such as water, oil or gas, and more particularly, to a solenoid valve applied to an automatic water flusher.

BACKGROUND OF THE INVENTION

(2) U.S. Pat. No. 10,208,870 discloses a solenoid valve. When the water is shut off, as shown in FIG. 4 and FIG. 5 of this patent, the sealing pad **34** is against the valve gate **26** inside the valve body **24**. Referring to FIG. 6 and FIG. 7 of this patent, when the coil A is energized, the electromagnetic field generated by the coil will move the iron core **31** upwards for the sealing pad

34 to leave the valve gate **26**, so that water can flow in via **12** the water inlet pipe **241** and flow out via the water outlet pipe **242**.

(3) However, as disclosed in U.S. Pat. No. 10,208,870, the direction of the piston **31** and the direction of the induced pressure are opposite, so they are against each other to occur water hammer, and the piston **31** is moved away by the force of water hammer. As a result, the water cannot be shut off effectively. The magnetic route of the magnet **14** passes through the magnetic shaft **33**, the piston **31**, the bottom cover **40** and the coil housing **10** from the upper half (N pole or S pole) of the magnet **14** and ends at the lower half of the magnetic shaft **33** (S pole or N pole). The bottom cover **40** and the coil housing **10** are made of a magnetically conductive material. Because the cross-section of the assembly of the bottom cover **40** and the coil housing **10** is in a C shape, the manufacturing cost and assembly time are increased. Besides, the magnet **14** is in contact with the magnetic shaft **33** to reduce the reluctance. The solenoid valve is unstable and the **24** efficiency is low. Accordingly, the inventor of the present invention has devoted himself based on his many years of practical experiences to solve these problems.

SUMMARY OF THE INVENTION

(4) The primary object of the present invention is to provide a solenoid valve, which has a bistable effect to form two magnetic routes of magnets.

(5) In order to achieve the above object, the present invention provides a solenoid valve. The solenoid valve comprises a water inlet valve seat, a guide seat, a coil barrel, a first magnet, a second magnet, and a water outlet valve seat. The water inlet valve seat includes a base. The base has a guide hole and at least one water inlet. One side of the base is formed with a first inner stop portion, an inner cylindrical portion, and an outer cylindrical portion. The inner cylindrical portion has a through hole. A flow passage is formed between the inner cylindrical portion and the outer cylindrical portion. The flow passage communicates with the through hole of the inner cylindrical portion. The outer cylindrical portion has a lower opening. The guide seat is disposed in the water inlet valve seat. The guide seat has a perforation. The guide seat includes a guide shaft, a guide spring, and a diaphragm. The guide shaft is arranged in the guide hole. One end of the guide spring abuts against the guide seat. The diaphragm is sleeved on the guide shaft and abuts against an inner surface of the inner cylindrical portion. The coil barrel is disposed in the water inlet valve seat. The coil barrel includes an axial tube, an upper seat, and a lower seat. Two ends of the axial tube are formed with a first portion and a second portion. The upper seat is arranged on an outer side of the first portion. Another end of the guide spring abuts against the upper seat. The lower seat is arranged on an outer side of the second portion. A coil portion is formed between the upper seat and the lower seat. At least one coil is coiled on the coil portion. The axial tube has an axial hole. A pressure guide block, an iron core spring, an iron core and a pressure relief block are provided in the axial hole. The pressure guide block has a water hole. The water hole communicates with the perforation. The iron core spring acts on the pressure guide block and the iron core. The iron core has an accommodating groove. A plug is provided in the accommodating groove. A pressure relief hole is formed in the pressure relief block. The first magnet is disposed in the pressure guide block or the upper seat. The second magnet is disposed in the pressure relief block or the lower seat. The first magnet and the second magnet form a magnetic route. The water outlet valve seat is disposed at the lower opening of the outer cylindrical portion. The water outlet valve seat has a water outlet. The water outlet communicates with the flow passage and the pressure relief hole.

(6) In the solenoid valve provided by the present invention, when in the open state, the iron core is moved towards the pressure guide block, and the plug leaves the pressure relief opening, so that water can flow out to reduce the pressure between the upper seat and the diaphragm. The guide seat and the diaphragm are moved towards the pressure guide block. Another water flow can flow out from the water inlet through the flow passages and the water outlet. When in the closed state, the iron core spring and water pressure push against the iron core to close the pressure relief hole, and

the diaphragm blocks the water flow in the water inlet via the water pressure and the force of the guide spring.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a perspective view of a preferred embodiment of the present invention;
- (2) FIG. 2 is an exploded view of the preferred embodiment of the present invention;
- (3) FIG. 3 is another exploded view of the preferred embodiment of the present invention;
- (4) FIG. 4 is a cross-sectional view of the preferred embodiment of the present invention;
- (5) FIG. 5 is a cross-sectional view taken along line 5-5 of FIG. 4;
- (6) FIG. 6 is a schematic view of the preferred embodiment of the present invention when in use, illustrating a closed state;
- (7) FIG. 7 is a schematic view of the preferred embodiment of the present invention when in use, illustrating an open state;
- (8) FIG. 8 is a cross-sectional view taken along line 8-8 of FIG. 7;
- (9) FIG. 9 is a cross-sectional view of another embodiment of the present invention, showing that the magnets are arranged in different positions; and
- (10) FIG. 10 is a cross-sectional view of a further embodiment of the present invention, showing that the magnets are arranged in different positions.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

- (11) Embodiments of the present invention will now be described, by way of example only, with reference to the accompanying drawings.
- (12) As shown in FIG. 1 through FIG. 4, the present invention discloses a solenoid valve **100**. The solenoid valve **100** is installed in a non-contact automatic water flusher, such as an infrared sensor faucet or a urinal tap. The automatic water flusher includes a sensor electrically connected to the solenoid valve **100**. The solenoid valve **100** comprises a water inlet valve seat **10**, a water-blocking cover **20**, a guide seat **30**, a coil barrel **40**, and a water outlet valve seat **50**.
- (13) The water inlet valve seat **10** is a hollow cylinder. Referring to FIG. 8, the water inlet valve seat **10** includes a base **11**, a first inner stop portion **12**, a second inner stop portion **13**, an inner cylindrical portion **14**, and an outer cylindrical portion **15**. The base **11** is a base plate. The base **11** has a guide hole **111** and a plurality of water inlets **112** surrounding the guide hole **111**. The guide hole **111** and the water inlets **112** are connected to a water source. The first inner stop portion **12**, the second inner stop portion **13**, the inner **12** cylindrical portion **14** and the outer cylindrical portion **15** are axially arranged on one side **113** of the base **11**, and are spaced apart from one other and arranged in a concentric circle in sequence outwardly from the water inlets **112**. As shown in FIG. 8, the first inner stop portion **12** and the second inner stop portion **13** are annular walls **421**. The inner cylindrical portion **14** and the outer cylindrical portion **15** are in a cylindrical shape. The second inner stop portion **13** and the inner cylindrical portion **14** each have a through hole **131**, **141**. The inner cylindrical portion **14** has an outer side provided with two arc-shaped protruding ribs **142** extending radially and outwardly. The protruding ribs **142** each have a wire hole **143**. The ribs **142** are connected to the outer cylindrical portion **15**, so that two flow passages **145** are formed between the inner cylindrical portion **14** and the outer cylindrical portion **15**. The flow passages **145** communicate with the through holes **131**, **141**, respectively. A lower opening **16** is formed on one end of the outer cylindrical portion **15** away from the base **11** of the water inlet valve seat **10**.
- (14) The water-blocking cover **20** covers a part of the base **11** and a part of the outer cylindrical portion **15** to form the flow passages **145**. In this embodiment, the outer cylindrical portion **15** and the base **11** are not closed. The flow passages **145** are formed by means of the water-blocking cover

20. Those skilled in the art can understand that the outer cylindrical portion **15** and the base **11** may be connected to each other to form the flow passages **145**, without using the water-blocking cover **20**.

(15) The guide seat **30** is disposed in the water inlet valve seat **10**. The guide seat **30** has a perforation **31**. The guide seat **30** includes a guide shaft **32**, a first guide portion **33**, a second guide portion **34** and a diaphragm **35**. One end of the guide shaft **32** is movably arranged in the guide hole **111**, so that the perforation **31** is connected to the water source. The first guide portion **33** and the second guide portion **34** are disposed on the other end of the guide shaft **32** and extend radially and outwardly. The first guide portion **33** and the second guide portion **34** are disc-shaped. The first guide portion **33** and the second guide portion **34** are spaced apart from each other. The first guide portion **33** is pressed against the base **11** in a closed position. The second guide portion **34** includes a guide ring **341** and a guide spring **342**. The guide ring **341** has two notches **343**. The guide spring **342** is located in the guide ring **341** and is pressed against the second guide portion **34**. The first guide portion **33** is pressed against the coil barrel **40** in an open position.

(16) The diaphragm **35** has a restricting portion **36**, a connecting portion **37** and a fixing portion **38** from inside to outside. The restricting portion **36** is substantially disc-shaped. The restricting portion **36** has an aperture **39**. The guide shaft **32** is inserted into the aperture **39**. The restricting portion **36** is located between the first guide portion **33** and the second guide portion **34**, so that the restricting portion **36** is linked with the guide seat **30**. When in the closed position, the restricting portion **36** is pushed by the second guide portion **34** to abut against the first inner stop portion **12**. The connecting portion **37** connects the restricting portion **36** and the fixing portion **38**. The cross-section of the connecting portion **37** is c-shaped or n-shaped. As shown in FIG. 4, when in the open position, the connecting portion **37** is pressed to move towards the coil barrel **40**. The fixing portion **38** is secured to the coil barrel **40** and abuts against the inner surface of the inner cylindrical portion **14**.

(17) The coil barrel **40** is disposed in the inner cylindrical portion **14** of the water inlet valve seat **10**. The coil barrel **40** includes an axial tube **41**, an upper seat **42**, and a lower seat **43**. Two ends of the axial tube **41** are formed with a first portion **411** and a second portion **412**. The upper seat **42** is arranged radially on the outer side of the first portion **411**. The upper seat **42** includes an annular wall **421** extending axially and outwardly. The annular wall **421** is formed with an opening **422**. The guide spring **342** is against the bottom of the opening **422**. A retaining recess **423** is formed on the outer side of the annular wall **421** for receiving the fixing portion **38**, so that only the opening **422** is in communication with the perforation **31**. The lower seat **43** is arranged radially on the outer side of the second portion **412**. A coil portion **44** is formed between the upper seat **42** and the lower seat **43**. At least one coil **441** is coiled on the coil portion **44**. The outer sides of the upper seat **42** and the lower seat **43** are respectively recessed to form a pair of rectangular grooves **424**, **431**. The rectangular grooves **424**, **431** are located at both ends in the radial direction. Circular holes **425**, **432** are defined in the bottoms of the rectangular grooves **424**, **431**. Cylindrical magnets **426**, **433** are provided in the circular holes **425**, **432**, respectively. Elongate magnetically conductive sheets **427** are connected between the rectangular grooves **424** of the upper seat **42** and the rectangular grooves **431** of the lower seat **43**, respectively. The magnetically conductive sheets **427** are located outside the coil **441** and the magnets **426**, **433**. Two ends of each of the magnetically conductive sheets **427** are connected to the magnets **426**, **433**, respectively. The magnets **426** of the upper seat **42** and the magnets **433** of the lower seat **43** are of opposite polarity. In this embodiment, the N poles of the magnets **426** of the upper seat **42** are directed towards the magnetically conductive sheet **427**, and the S poles of the magnets **433** of the lower seat **43** are directed towards the other magnetically conductive sheet **427**, so as to enhance the magnetic force. The magnetically conductive sheets **427** are not essential components in the invention. Two pairs of magnets **426**, **433** form two complete magnetic routes. The magnetically conductive sheets **427** are configured to strengthen the magnetic force. The solenoid valve **100** has a simple structure and can

be produced easily. The guide seat **30** is made of a non-magnetic material, so it is able to reduce magnetic field interference.

(18) The axial tube **41** has an axial hole **45** passing through both ends thereof. A pressure guide block **46**, an iron core **47** and a pressure relief block **48** are provided in the axial hole **45**. The pressure guide block **46** is made of a magnetically conductive material. The pressure guide block **46** is screwed to the axial hole **45** and located at the first portion **411**. The pressure guide block **46** is against the other end of the guide spring **342**. The pressure guide block **46** has a pair of water holes **461**. The water holes **461** of the pressure guide block **46** communicate with the opening **422**. The iron core **47** is located between the pressure guide block **46** and the pressure relief block **48**. The outer diameter of the iron core **47** is slightly smaller than the inner diameter of the axial hole **45**, so that a gap is formed between the iron core **47** and the axial hole **45** for water to pass there through. As shown in FIG. 5, one end of the iron core **47**, facing the pressure guide block **46**, has a receiving groove **471**. The receiving groove **471** communicates with the water holes **461** of the pressure guide block **46**. The receiving groove **471** is configured to receive an iron core spring **472**. Two ends of the iron core spring **472** are against the pressure guide block **46** and the iron core **47**, respectively. The other end of the iron core **47** has an accommodating groove **473**. The wall of the accommodating groove **473** is formed with two notches **474**. A plug **475** is provided in the accommodating groove **473**. The plug **475** includes an inner iron core covered with an outer rubber layer. The outer side of the plug **475** and the bottom of the notch **474** are on the same plane. The pressure relief block **48** is screwed to the axial hole **45** and located at the second portion **412**. A pressure relief hole **481** is formed in the pressure relief block **48**. One end of the pressure relief block **48** is formed with a tapered pressure relief opening **482**. When in the closed position, the iron core spring **472** enables the wall of the accommodating groove **473** to be pressed against the pressure relief block **48**, so as to avoid excessive compression of the plug **475**. The plug **475** is pressed against the pressure relief opening **482** to close the pressure relief hole **481**. One end of the pressure relief block **48** is formed with a linear adjustment groove **483**. The outer wall of the pressure relief block **48** is formed with an external thread, and the axial hole **45** of the second portion **412** is formed with an internal thread, so that the gap between the pressure relief block **48** and the iron core **47** can be adjusted.

(19) The water outlet valve seat **50** is located at the lower opening **16** of the water inlet valve seat **10**. The water outlet valve seat **50** has a water outlet **51**. One end of the water outlet **51** is in communication with the flow passages **145** and the pressure relief hole **481**. The axial tube **41** extends out of the lower seat **43** and is located in the axial hole **45**. The other end of the water outlet **51** may be connected to the automatic water flusher.

(20) Referring to FIG. 6, when in use, the tap water of the water source will flow into both the perforation **31** and the water inlets **112**. The first water flow passes through the perforation **31**, the opening **422** and the water inlets **461** to enter the axial hole **45**. The water flow and the elastic force of the iron core spring **472** will push against the iron core **47**, so that the iron core **47** is pushed towards the pressure relief block **48** to close the pressure relief hole **481**. The pressure of the opening **422** and the elastic force of the guide spring **342** will push the guide seat **30** and the diaphragm **35** towards the base **11**, so that the restricting portion **36** of the diaphragm **35** will be against the first inner stop portion **12**. The second water flow passes through the water inlets **112** to enter the water inlet valve seat **10**. Because the sum of the pressure of the opening **422** and the elastic force of the guide spring **342** is greater than the pressure of the second water flow, the second water flow will be blocked by the diaphragm **35** to keep the solenoid valve **100** in a closed state.

(21) Referring to FIG. 7, the input current to the coil **441** generates a positive incremental magnetic field in the same direction as the magnetic force lines of the magnets. The iron core **47** and the pressure guide block **46** generate a magnetic attraction force, so that the iron core **47** is moved towards the pressure guide block **46**. The core spring **472** is compressed. When the plug **475** leaves

the pressure relief opening **482**, the first water flow can flow out via the gap, the pressure relief opening **482**, the pressure relief hole **481** and the water outlet **51** to reduce the pressure of the opening **422**. At this time, the sum of the pressure of the opening **422** and the elastic force of the guide spring **342** is less than the pressure of the second water flow. The pressure of the second water flow will push the guide seat **30** and the diaphragm **35** towards the pressure guide block **46** to compress the guide spring **342**, so that the water inlets **112**, the through hole **131** of the second inner stop portion **13**, the through hole **141** of the inner cylindrical portion **14**, and the flow passages **145** are in communication with one another. The second water flow can flow out via the through hole **131** of the second inner stop portion **13**, the through hole **141** of the inner cylindrical portion **14**, the flow passages **145** and the water outlet **51**. The second water flow flows into the automatic water flusher together with the first water flow. After the current is cut off, the magnetic routes of the magnets **426**, **433** and the magnetically conductive sheets **427** keep the solenoid valve **100** in an open state, thereby saving electric energy.

(22) Referring to FIG. **6**, when the input current to the coil **441** generates a reverse magnetic field, the iron core **47** and the pressure guide block **46** reduce attraction, so that the iron core **47** is moved towards the pressure relief block **48** to close the pressure relief hole **481** and increase the pressure of the opening **422**. The guide seat **30** and the diaphragm **35** are moved towards the first inner stop portion **12**, and the diaphragm **35** abuts against the first inner stop portion **12** to block the second water flow, so as to shut off the water flow. After the current is cut off, through the elastic force of the iron core spring **472**, the water pressure of the first water flow is simultaneously introduced to push against the iron core **47** to keep the pressure relief hole **481** closed, and the pressure of the opening **422** and the elastic force of the guide spring **342** block the second water flow to achieve a bistable effect. When water hammer occurs, water hammer will further push against the iron core **47** to increase the force to close the pressure relief hole **481**. Simultaneously, the sum of the pressure of the opening **422** and the elastic force of the guide spring **342** is still greater than the pressure of the second water flow, thereby maintaining the closed state, so as to achieve the effect of a bistable solenoid valve.

(23) In the present invention, the magnetic routes are formed by two pairs of magnets. The number of the magnets are not limited to two pairs, which may be one pair or more than two pairs. A magnetic route is formed by a single magnet **426** and a single magnet **433** that are located on the same side. The single magnet **426** and the single magnet **433** may be arranged in different positions. The magnet **426** may be disposed in the pressure guide block **46** or in the circular hole **425** of the upper seat **42**. The circular hole **425** may be located close to the pressure guide block **46** or the iron core **47**. The magnet **433** may be disposed in the pressure relief block **48** or in the circular hole **432** of the lower seat **43**. The circular hole **432** may be located close to the pressure relief block **48** or the iron core **47**. Referring to FIG. **9**, the pressure guide block **46**, the pressure relief block **48** or both are made of a magnetically conductive material. The magnet **426** is disposed in the pressure guide block **46**. The magnet **426** has a first hole **428** therein. The magnet **433** is disposed in the pressure relief block **48**. The magnet **433** has a second hole **434** therein. Referring to FIG. **10**, the pressure guide block **46** is made of a non-magnetically conductive material. The pressure relief block **48** is made of a magnetically conductive material. The magnet **426** is disposed in the circular hole **425** of the upper seat **42**. The circular hole **425** is located close to the end of the iron core **47**. The magnet **433** is disposed in the circular hole **432** of the lower seat **43**. The circular hole **432** is located close to the pressure relief block **48**.

(24) Although particular embodiments of the present invention have been described in detail for purposes of illustration, various modifications and enhancements may be made without departing from the spirit and scope of the present invention. Accordingly, the present invention is not to be limited except as by the appended claims.

Claims

1. A solenoid valve, comprising: a water inlet valve seat, including a base, the base having a guide hole and at least one water inlet, one side of the base being formed with a first inner stop portion, an inner cylindrical portion and an outer cylindrical portion, the inner cylindrical portion having a through hole, a flow passage being formed between the inner cylindrical portion and the outer cylindrical portion, the flow passage communicating with the through hole of the inner cylindrical portion, the outer cylindrical portion having a lower opening; a guide seat, disposed in the water inlet valve seat, the guide seat having a perforation, the guide seat including a guide shaft, a guide spring and a diaphragm, the guide shaft being arranged in the guide hole, one end of the guide spring abutting against the guide seat, the diaphragm being sleeved on the guide shaft and abutting against an inner surface of the inner cylindrical portion; a coil barrel, disposed in the water inlet valve seat, the coil barrel including an axial tube, an upper seat and a lower seat, two ends of the axial tube being formed with a first portion and a second portion, the upper seat being arranged on an outer side of the first portion, another end of the guide spring abutting against the upper seat, the lower seat being arranged on an outer side of the second portion, a coil portion being formed between the upper seat and the lower seat, at least one coil being coiled on the coil portion; the axial tube having an axial hole, a pressure guide block, an iron core spring, an iron core and a pressure relief block being provided in the axial hole, the pressure guide block having a water hole, the water hole communicating with the perforation, the iron core spring acting on the pressure guide block and the iron core, the iron core having an accommodating groove, a plug being provided in the accommodating groove, a pressure relief hole being formed in the pressure relief block; a first magnet and a second magnet, the first magnet being disposed in one of the pressure guide block and the upper seat, the second magnet being disposed in one of the pressure relief block and the lower seat, the first magnet and the second magnet forming a magnetic route; a water outlet valve seat, disposed at the lower opening of the outer cylindrical portion, the water outlet valve seat having a water outlet, the water outlet communicating with the flow passage and the pressure relief hole.

2. The solenoid valve as claimed in claim 1, wherein the pressure guide block is screwed to the axial hole and located at the first portion, the pressure relief block is screwed to the axial hole and located at the second portion, the iron core is located between the pressure guide block and the pressure relief block; the solenoid valve has an open state and a closed state; when in the open state, the iron core is moved towards the pressure guide block for the water hole to communicate with the pressure relief hole, and the guide seat and the diaphragm are moved towards the pressure guide block for the water inlet to communicate with the through hole of the inner cylindrical portion; when in the closed state, the iron core is moved towards the pressure relief block, the plug abuts against the pressure relief hole of the pressure relief block, and the diaphragm abuts against the first inner stop portion, so that the water inlet is not in communication with the through hole of the inner cylindrical portion.

3. The solenoid valve as claimed in claim 1, wherein the coil barrel includes a pair of magnetically conductive sheets, and the magnetically conductive sheets are connected to the magnets of the upper seat and the magnets of the lower seat, respectively.

4. The solenoid valve as claimed in claim 1, wherein the inner cylindrical portion has an outer side provided with at least one protruding rib extending radially and outwardly, the protruding rib has a wire hole, and the rib is connected to the outer cylindrical portion so that the flow passage is formed between the inner cylindrical portion and the outer cylindrical portion.

5. The solenoid valve as claimed in claim 1, wherein the guide seat includes a first guide portion and a second guide portion, the first guide portion and the second guide portion are spaced apart from each other, the diaphragm includes a restricting portion, a connecting portion and a fixing

portion, the restricting portion is arranged on the guide seat and located between the first guide portion and the second guide portion, the connecting portion connects the restricting portion and the fixing portion, and the fixing portion is arranged on the upper seat.

6. The solenoid valve as claimed in claim 1, wherein the iron core has an outer diameter slightly smaller than an inner diameter of the axial hole.

7. The solenoid valve as claimed in claim 1, wherein the iron core has a receiving groove, the receiving groove communicates with the water hole of the pressure guide block, and the receiving groove is configured to receive the iron core spring.

8. The solenoid valve as claimed in claim 1, wherein a wall of the accommodating groove is formed with two notches.

9. The solenoid valve as claimed in claim 1, wherein the pressure relief block is screwed to the axial hole, and the pressure relief block is formed with an adjustment groove.

10. A solenoid valve, comprising: a water inlet valve seat, having a water inlet and a lower opening; a coil barrel, disposed in the water inlet valve seat, the coil barrel including an axial tube, an upper seat and a lower seat, two ends of the axial tube being formed with a first portion and a second portion, the upper seat being arranged on an outer side of the first portion, the lower seat being arranged on an outer side of the second portion, a coil portion being formed between the upper seat and the lower seat, at least one coil being coiled on the coil portion; the axial tube having an axial hole, a pressure guide block, an iron core spring, an iron core and a pressure relief block being provided in the axial hole, the pressure guide block having a water hole, the water inlet communicating with the water hole, the iron core spring acting on the pressure guide block and the iron core, the iron core having an accommodating groove, a plug being provided in the accommodating groove, a pressure relief hole being formed in the pressure relief block; a first magnet and a second magnet, the first magnet being disposed in one of the pressure guide block and the upper seat, the second magnet being disposed in one of the pressure relief block and the lower seat, the first magnet and the second magnet forming a magnetic route; a water outlet valve seat, disposed at the lower opening, the water outlet valve seat having a water outlet, the water outlet communicating with the pressure relief hole.

11. The solenoid valve as claimed in claim 10, wherein the pressure guide block is screwed to the axial hole and located at the first portion, the pressure relief block is screwed to the axial hole and located at the second portion, the iron core is located between the pressure guide block and the pressure relief block; the solenoid valve has an open state and a closed state; when in the open state, the iron core is moved towards the pressure guide block for the water hole to communicate with the pressure relief hole; when in the closed state, the iron core is moved towards the pressure relief block, and the plug abuts against the pressure relief hole of the pressure relief block.

12. The solenoid valve as claimed in claim 10, wherein the coil barrel includes a pair of magnetically conductive sheets, and the magnetically conductive sheets are connected to the magnets of the upper seat and the magnets of the lower seat, respectively.
